

Manjeet Tiwari

Backend Engineer · Java / Spring Boot

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SUMMARY

Final-year B.Tech Computer Science student (2027) focused on backend software engineering in Java and Spring Boot — REST APIs, PostgreSQL, and correctness under retries and concurrency. Built transactional money movement with a double-entry ledger and outbox, JWT role-based authorization, and hybrid-encryption payment simulation.

EDUCATION

Guru Gobind Singh Indraprastha University, New Delhi, India

2023 – Expected June 2027

B.Tech, Computer Science & Technology · GPA 8.9/10 overall · 9.23/10 (6th semester)

Class 12

82.8% | March 2023

Class 10

93.16% | March 2021

TECHNICAL SKILLS

Languages: Java, TypeScript, SQL, Python, JavaScript (fundamentals)

Backend: Spring Boot, Spring Security, JPA/Hibernate, REST APIs, JWT, NestJS, TypeORM

Data & Messaging: PostgreSQL, MySQL, H2, Flyway, Apache Kafka

Platform & Testing: Docker, Docker Compose, GitHub Actions, Git, JUnit, MockMvc, Testcontainers

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Familiar: Redis, MongoDB, FastAPI, Kubernetes, SonarQube, AWS (EC2)

EXPERIENCE

AI & Generative AI Intern — IBM SkillsBuild (Edunet Foundation)

Aug 2026 – Present

Generative AI, Agentic AI, Machine Learning, IBM watsonx, watsonx Orchestrate

- Studying machine learning foundations and generative AI: large language models, prompt design, and responsible-use limits.
- Composing agents from tools and skills in IBM watsonx Orchestrate to automate multi-step workflows rather than one-shot prompts.

Backend Developer Intern — PearlThoughts

Jul 2026 – Aug 2026

NestJS, TypeScript, PostgreSQL, TypeORM, REST APIs, JWT

- Built *Schedula*, a doctor–patient appointment backend: JWT auth, onboarding, recurring availability with date overrides.
- Implemented *STREAM* and *WAVE* booking modes on PostgreSQL with pessimistic locking, a 30-minute cancel/reschedule cutoff, and next-available-slot suggestions.

PROJECTS

SAAKH — Financial Transfer & Reconciliation Backend

[Source](#)

Java 21, Spring Boot 3.5, PostgreSQL, Flyway, JPA/Hibernate, Kafka, Docker, Testcontainers (47 tests), GitHub Actions

- Posted each transfer in one PostgreSQL transaction — debit/credit, transfer row, immutable double-entry ledger, idempotency record, unpublished outbox event; Kafka never sits on the money path.
- Locked accounts with ordered SELECT ... FOR UPDATE, removing lost updates and A→B / B→A deadlocks under concurrent transfers.
- Enforced idempotency with an Idempotency-Key plus SHA-256 body fingerprint: identical replays return the original transfer, conflicting reuse is rejected with 409.
- Published events only after commit via a transactional outbox (SKIP LOCKED, stable event IDs, transferId partition key, retries) at-least-once, with detect-only reconciliation.

UPI Offline Mesh Payment System — deferred settlement (simulation)

[Source](#)

Java, Spring Boot, RSA-OAEP, AES-256-GCM, SHA-256, H2

- Simulated offline payments as encrypted packets hopping a software mesh of untrusted devices until an internet-connected bridge uploads them for settlement (not NPCI/UPI infrastructure).
- Protected packets with hybrid RSA-OAEP + AES-256-GCM, GCM tamper detection, signedAt freshness windows, and SHA-256 hashing.
- Blocked double settlement with an atomic hash claim and a unique packet-hash constraint; duplicates and replays are rejected.

Common Login Module — JWT Authentication & Role-Based Access Control

Java, Spring Boot, Spring Security, JWT, JPA/Hibernate, BCrypt, H2

- Built a reusable login module: JwtAuthenticationFilter, CustomUserDetailsService, BCrypt hashing, SecurityContextHolder.
- Separated authentication from authorization using USER / MODERATOR / ADMIN roles, method-level @PreAuthorize (including admin-or-self), and global JSON 401/403/validation handling.

ACHIEVEMENTS & CERTIFICATIONS

- Microsoft TechJam 2.0 — advanced to Round 3 with Team S-Squad, Microsoft Sovereign Office, Noida (Nov 2025); Smart India Hackathon 2025 participant, Team quantum_Coders (Sep 2025).
- Certifications: Spring 5 Basics with Spring Boot and Spring Microservices — Infosys Springboard (Apr 2026); Web Development Professional — MTF Institute (Jun 2026).